PEOPLEOPS DATA HACKATHON · INDUS WEAVE TEXTILES

HR Pipeline & Overtime Analysis

Presented by Muhammad Raza

Auditing payroll disbursements against biometric punch logs to uncover overtime leakage, sensor calibration errors, and attendance discrepancies.



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DIAGNOSTIC & REMEDIATION

Problem & Solution

Reconciling authorized overtime requests with physical biometric capture through an automated data pipeline.

The Problem

Payroll calculates and pays overtime based strictly on approved requests, but actual working time is tracked via biometric punches. **The numbers do not match.**

Disconnection between manual approval workflows and physical access logs creates systemic payment leakage.

The Solution

An automated **6-stage Python + SQL pipeline** designed to ingest, validate, and compute actual overtime across production units.

Target Architecture: Raw Data → Staging → Data Warehouse (Star Schema) → Verified Operational Answers.

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FINDING 01 · FINANCIAL RECONCILIATION

Finding 1: Overtime Leakage

Departments are disbursing massive overtime payments when compared to actual computed hours from physical attendance.

405 vs 10

Paid overtime hours vs. biometric computed hours in Human Resources.

Metric (HR Dept)	Hours	Status
Computed Biometric OT	10 hrs	Verified
Payroll Disbursed OT	405 hrs	Discrepancy
Net Unverified Leakage	+395 hrs	Actionable

Recommended Action: Link payroll authorization directly to biometric-computed overtime hours before monthly disbursement.

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FINDINGS 02 & 03 · AUDIT EXCEPTIONS

Findings 2 & 3: Data Integrity

Category	Observed Issue	Required Action
Faulty Device DEV_05	Hardware clock was running **1 hour ahead** of standard time, corrupting punch timestamps and invalidating computed shifts.	**Immediate IT Maintenance** Resync NTP daemon and audit affected historical shifts.
Ghost Employees Weaving & Cutting	Identified employees received full salaries in September 2026 despite recording **zero biometric attendance**.	**Freeze Payroll** Halt automated disbursement pending HR compliance investigation.

Integrity audits run automatically during pipeline Stage 2 and Stage 4 ingest validation.

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STACK & PIPELINE ARCHITECTURE

Architecture & Quality Controls

01 **Tech Stack.** Python, SQLAlchemy ORM, and Microsoft SQL Server.

Built on an analytical Star Schema optimized for departmental overtime aggregation.

02 **Data Quality.** Automated verification suites.

Enforces Primary Key uniqueness and foreign key referential integrity at every batch ingest.

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Thank you!

Questions & Discussion

Ready for deployment & ERP integration.

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